

A prop test 'graphical poem' addressing the question:

For survey of 161 individuals on willingness to serve as organ donors in the case of an accident, is there evidence that responses rate differs from a 50/50 split, when 53 individuals respond 'no' and 108 individuals respond 'yes'?

1. Draw bar chart (stacks) for each category.
2. Draw theoretical limits for proportion (0 to 1).
3. Calculate proportion and place (also the balancing point for stacks!).
4. Place null (look at the initial question, 'differs from 50/50' in this case)
5. Draw distribution for NULL, where $SD = \sqrt{(\pi * (1-\pi))/n}$
6. Calc z-score – where does observed prop fit into NULL distributions (i.e. how many standard deviations fit between .5 and .67)

